

$$h_{s1} = L \sin \theta \quad \frac{\partial^2 h_{s1}}{\partial \theta^2} = -L \sin \theta$$

$$K_{s1} = m_1 g \begin{bmatrix} -L \sin \theta_0 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 h_{s2}}{\partial \theta^2} = -(L_0 - R \varphi) \sin \theta - R \cos \theta$$

$$\frac{\partial^2 h_{s2}}{\partial \theta \partial \varphi} = -R \cos \theta$$

$$\frac{\partial^2 h_{s2}}{\partial \varphi^2} = 0$$

$$K_{s2} = m_2 g \begin{bmatrix} -L_0 \sin \theta_0 & -R \cos \theta_0 & -R \cos \theta_0 \\ -R \cos \theta_0 & 0 & 0 \end{bmatrix}$$

$$\frac{\partial^2 \Delta L_1}{\partial \theta^2} = -2L \sin \theta$$

$$K_{EL1, II} = k_1 \Delta L_1 \begin{bmatrix} -2L \sin \theta_0 & 0 \\ 0 & 0 \end{bmatrix}$$

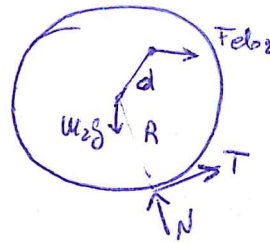
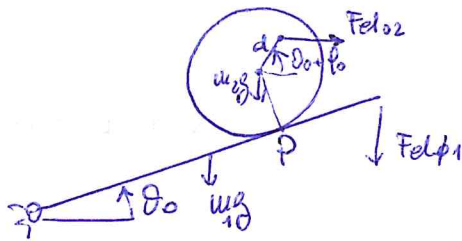
$$\frac{\partial^2 \Delta L_2}{\partial \theta^2} = (L_0 - R \varphi) \cos \theta - R \sin \theta + d \cos(\theta + \varphi_0 + \varphi)$$

$$\frac{\partial^2 \Delta L_2}{\partial \theta \partial \varphi} = -R \sin \theta + d \cos(\theta + \varphi_0 + \varphi)$$

$$\frac{\partial^2 \Delta L_2}{\partial \varphi^2} = d \cos(\theta + \varphi_0 + \varphi)$$

$$\cancel{K_{EL2, II} = k_2 \Delta L_2 \begin{bmatrix} L_0 \cos \theta_0 - R \sin \theta_0 + d \cos(\theta_0 + \varphi_0) & -R \sin \theta_0 + d \cos(\theta_0 + \varphi_0) \\ -R \sin \theta_0 + d \cos(\theta_0 + \varphi_0) & d \cos(\theta_0 + \varphi_0) \end{bmatrix}}$$

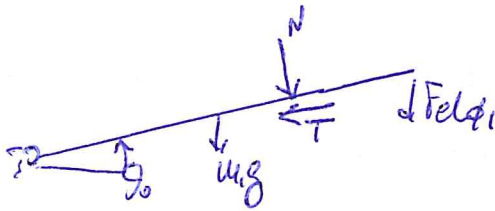
$$K_{EL2, II} = k_2 \Delta L_2 \begin{bmatrix} L_0 \cos \theta_0 - R \sin \theta_0 + d \cos(\theta_0 + \varphi_0) & -R \sin \theta_0 + d \cos(\theta_0 + \varphi_0) \\ -R \sin \theta_0 + d \cos(\theta_0 + \varphi_0) & d \cos(\theta_0 + \varphi_0) \end{bmatrix}$$



$$w_2 g R \sin \theta_0 = F_{el2} (R \cos \theta_0 + d \sin(\theta_0 + \phi_0))$$

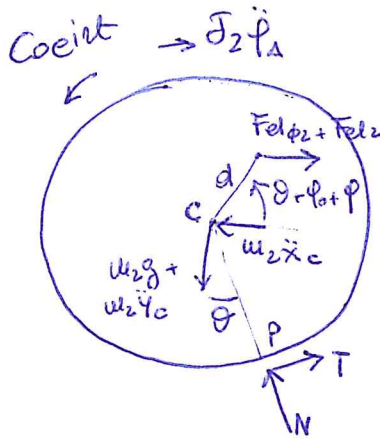
$$F_{el2} = w_2 g \left(\frac{R \cos \theta_0 + d \sin(\theta_0 + \phi_0)}{R \sin \theta_0} \right)$$

$$N = w_2 g \cos \theta_0 + F_{el2} \sin \theta_0$$



$$F_{el1} 2L \cos \theta_0 + w_1 g L \cos \theta_0 + N L_0 = 0$$

$$F_{el1} = - \frac{w_1 g L \cos \theta_0 + N L_0}{2L \cos \theta_0}$$



$$TR + \text{Coint} - \ddot{\theta}_2 \dot{\phi}_A - (F_{el1} + F_{el2}) d \sin(\theta_0 + \phi_0) = 0$$

$$TR + \text{Coint} - \ddot{\theta}_2 \dot{\phi}_A - F_{el2} d \sin(\theta_0 + \phi_0) - F_{el1} d \cos(\theta_0 + \phi_0) (\dot{\theta} - \dot{\theta}_0) = 0$$

$$-F_{el1} d \cos(\theta_0 + \phi_0) \dot{\phi} - F_{el2} d \sin(\theta_0 + \phi_0) \dot{\phi} = 0$$

$$T_{ST} = \frac{F_{el2} d \sin(\theta_0 + \phi_0)}{R}$$

$$T_{dyn} = \frac{\ddot{\theta}_2 \dot{\phi}_A - \text{Coint} + F_{el2} d \cos(\theta_0 + \phi_0) (\dot{\theta} - \dot{\theta}_0 + \dot{\phi}) + F_{el2} d \sin(\theta_0 + \phi_0)}{R}$$

$$N \cos \theta - w_2 g - w_2 \ddot{\gamma}_c + T \sin \theta = 0$$

$$N \cos \theta_0 - w_2 g - w_2 \ddot{\gamma}_c + T_{ST} \cos \theta_0 (\theta - \theta_0) + T_{dyn} \sin \theta_0 = 0$$

$$N_{dyn} = \frac{w_2 g + w_2 \ddot{\gamma}_c - T_{ST} \cos \theta_0 (\theta - \theta_0) - T_{dyn} \sin \theta_0}{\cos \theta_0}$$

$$N_{ST} = \frac{w_2 g - T_{ST} \sin \theta_0}{\cos \theta_0}$$

$$N_{ST} - \frac{N_{dyn}}{\gamma_0} \Big|_{\omega_{max}} \gamma_0 = 0 \rightarrow \gamma_0 = \frac{N_{ST}}{\frac{N_{dyn}}{\gamma_0} \Big|_{\omega_{max}}}$$